

# First overview of governing equations

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## 1 A note on notations

### 1.1 Material derivative

The following equations are all equivalent, which can be shown algebraically.

$$\begin{aligned}\frac{d}{dt} &= \frac{\partial}{\partial t} + u \frac{\partial}{\partial x} + v \frac{\partial}{\partial y} + w \frac{\partial}{\partial z} \\ &= \frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla \\ &= \frac{\partial}{\partial t} + \mathbf{u} \cdot \text{div}()\end{aligned}$$

### 1.2 Tensors

$$\tau_{ij} \frac{\partial u_i}{\partial x_j} = \boldsymbol{\tau} : \nabla \mathbf{u}$$

where

$$\nabla \mathbf{u} = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{pmatrix}$$

The  $:$  is called the double contraction operator or the matrix inner product. It contracts two tensors into a scalar. Since  $\tau$  is the symmetric stress tensor,

## 2 Starting equations (partial differential equations)

### 2.1 Energy equation

#### 2.1.1 CRB

CRB: Cushman-Roisin and Beckers, 2011

$$\rho C_v \frac{dT}{dt} + p \left( \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right) = k_T \nabla^2 T \quad (1)$$

I refer to Schubert et al., 2001 for the conversion into the constant-pressure heat capacity variant of this equation.

#### 2.1.2 Jarvis & McKenzie, 1980

jamc80 Jarvis and McKenzie, 1980:

$$\rho C_p \left( \frac{\partial T}{\partial t} - \frac{\alpha T}{\rho C_p} \frac{\partial P}{\partial t} + \mathbf{u} \cdot [\nabla T - (\nabla T)_s] \right) = \nabla \cdot (K \nabla T) + H + \tau_{ij} \frac{\partial u_i}{\partial x_j} \quad (2)$$

### 2.1.3 Gassmüller et al., 2020

Gassmüller: Gassmüller et al., 2020

$$\rho C_p \left( \frac{\partial T}{\partial t} + \mathbf{u} \cdot \nabla T \right) - \nabla \cdot (k \nabla T) = \rho H + \tau : \dot{\varepsilon} + \alpha T (\mathbf{u} \cdot \nabla p) + Q_L \quad (3)$$

### 2.1.4 Schubert et al., 2001

Schubert et al., 2001 presents three thermmal energy equations:

The general form of the energy equation (6.9.1) is:

$$\rho T \frac{Ds}{Dt} = \tau_{ij} \frac{\partial u_i}{\partial x_j} + \frac{\partial}{\partial x_j} \left( k \frac{\partial T}{\partial x_j} \right) + \rho H \quad (4)$$

For an incompressible Newtonian fluid (eq. 6.9.13 + 6.9.14):

$$\rho C_v \frac{DT}{Dt} = \frac{\partial}{\partial x_i} + \frac{\mu}{2} \left( \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^2 + \rho H \quad (5)$$

with  $\frac{\mu}{2} \left( \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^2$  the viscous dissipation function.

In terms of  $C_p$  instead of  $C_v$  (for a compressible fluid) (eq. 6.9.16):

$$\rho C_p \left( \frac{DT}{Dt} - \left( \frac{\partial T}{\partial p} \right)_s \frac{Dp}{Dt} \right) = \frac{\partial}{\partial x_i} + \frac{\mu}{2} \left( \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^2 + \rho H \quad (6)$$

with (eq. 6.9.15, 6.9.17):

$$\begin{aligned} \left( \frac{\partial T}{\partial p} \right)_s &= \frac{\alpha T}{\rho C_p} \\ \left( \frac{DT}{Dt} \right)_s &= \left( \frac{\partial T}{\partial p} \right)_s \frac{Dp}{Dt} = \frac{\alpha T}{\rho C_p} \frac{Dp}{Dt} \end{aligned}$$

## 2.2 Momentum equation

### 2.2.1 CRB

CRB: Cushman-Roisin and Beckers, 2011 per component

$$x : \rho \left( \frac{du}{dt} + f_* w - f v \right) = -\frac{\partial p}{\partial x} + \frac{\partial \tau^{xx}}{\partial x} + \frac{\partial \tau^{xy}}{\partial y} + \frac{\partial \tau^{xz}}{\partial z} \quad (7a)$$

$$y : \rho \left( \frac{dv}{dt} + f u \right) = -\frac{\partial p}{\partial y} + \frac{\tau^{xy}}{\partial x} + \frac{\partial \tau^{yy}}{\partial y} + \frac{\partial \tau^{yz}}{\partial z} \quad (7b)$$

$$z : \rho \left( \frac{dw}{dt} - f_* u \right) = -\frac{\partial p}{\partial z} - \rho g + \frac{\tau^{xz}}{\partial x} + \frac{\tau^{yz}}{\partial y} + \frac{\partial \tau^{zz}}{\partial z} \quad (7c)$$

### 2.2.2 Jarvis & McKenzie, 1980

jamc80, eq. 4 Jarvis and Mckenzie, 1980:

$$\rho \left( \frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla P + \rho \mathbf{g} + \frac{\partial}{\partial x_j} \tau_{ij} \quad (8)$$

### 2.2.3 Gassmüller et al., 2020

Gassmüller Gassmüller et al., 2020:

$$-\nabla \cdot \tau + \nabla p = \rho \mathbf{g} \quad (9)$$

### 2.2.4 Schubert et al. 2001

$$\rho \frac{du_i}{dt} = -\frac{\partial p}{\partial x_i} + \frac{\partial \tau_{ij}}{\partial x_j} + \rho g_i \quad (10)$$

In 2D, the momentum equation components are:

$$\rho \left( \frac{\partial u_x}{\partial t} + u_x \frac{\partial u_x}{\partial x} + u_y \frac{\partial u_x}{\partial y} \right) = -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y}, \quad (11)$$

$$\rho \left( \frac{\partial u_y}{\partial t} + u_x \frac{\partial u_y}{\partial x} + u_y \frac{\partial u_y}{\partial y} \right) = -\frac{\partial p}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \rho g. \quad (12)$$

Note that the assumption is made that gravity acts in the positive y-direction. Section 6.14?

## 2.3 Mass equation (continuity equation) (trivial)

jamc80, eq. 3 Jarvis and Mckenzie, 1980:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (13)$$

CRB Cushman-Roisin and Beckers, 2011:

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w) = 0 \quad (14)$$

Gassm  ller Gassm  ller et al., 2020:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (15)$$

schubert et al. (2001) Schubert et al., 2001

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_1}(\rho u_i) = \frac{1}{\rho} \frac{D\rho}{Dt} + \frac{\partial u_1}{\partial x_1} = 0 \quad (16)$$

where  $\frac{D}{Dt}$  represents the material derivative (1.1). For incompressible flow,  $\frac{D\rho}{Dt} = 0$ , so eq 16 reduces to:

$$\frac{\partial u_i}{\partial x_i} = \nabla \cdot \mathbf{u} = 0 \quad (17)$$

## 2.4 Starting equations (constitutive equations)

## 2.5 Initial and boundary conditions

# 3 Appendix 1: tables of constants and reference/scaling values used

## References

- Cushman-Roisin, B., & Beckers, J.-M. (2011). *Introduction to geophysical fluid dynamics: Physical and numerical aspects*. Academic Press.
- Gassm  ller, R., Dannberg, J., Bangerth, W., Heister, T., & Myhill, R. (2020). On formulations of compressible mantle convection. *Geophysical Journal International*, 221(2), 1264–1280. <https://doi.org/10.1093/gji/ggaa078>
- Jarvis, G. T., & Mckenzie, D. P. (1980). Convection in a compressible fluid with infinite prandtl number. *Journal of Fluid Mechanics*, 96(3), 515–583. <https://doi.org/10.1017/S002211208000225X>
- Schubert, G., Turcotte, D. L., & Olson, P. (2001). *Mantle convection in the earth and planets*. Cambridge University Press.

Table 1: Constants used by Schubert et al., 2001

| Constant | Value | Name                                           |
|----------|-------|------------------------------------------------|
| $k_b$    | 0     | Bulk viscosity                                 |
| $\alpha$ |       | thermal expansivity                            |
| $\chi_T$ |       | isothermal compressibility                     |
| $c_v$    |       | specific heat at constant volume               |
| $c_p$    |       | specific heat at constant pressure             |
| $\chi_a$ |       | adiabatic compressibility                      |
| $k$      |       | thermal conductivity                           |
| $H$      |       | rate of internal heat production per unit mass |
| $K_a$    |       | adiabatic bulk modulus                         |
| $K_T$    |       | isothermal bulk modulus                        |

Table 2: Reference values used by Schubert et al., 2001

| Symbol       | Value                                 | Description                                                      |
|--------------|---------------------------------------|------------------------------------------------------------------|
| $\Delta T$   |                                       | Characteristic temperature difference driving thermal convection |
| $\chi_{T_r}$ |                                       | Representative isothermal compressibility                        |
| $\alpha_r$   |                                       | Representative thermal expansivity                               |
| $u_r$        | $\frac{k_r}{\rho_r c_{p_r} b}$        | Velocity scale                                                   |
| $\mu_r$      | $10^{21}$ Pa s                        | Representative viscosity                                         |
| $\chi_a$     |                                       | Adiabatic compressibility                                        |
| $b$          |                                       | Depth of convecting region                                       |
| $p_r$        | $\frac{\mu_r u_r}{b}$                 | Scale for dynamic pressure variations                            |
| $k_r$        | $4 \text{ W m}^{-1} \text{ K}^{-1}$   | Reference value thermal conductivity                             |
| $c_{p_r}$    | $1 \text{ kJ kg}^{-1} \text{ K}^{-1}$ | Reference specific heat at constant pressure                     |

Table 3: Dimensionless parameters and their values for the Earth's mantle

| Parameter  | Definition                                      | Value                | Description                                             |
|------------|-------------------------------------------------|----------------------|---------------------------------------------------------|
| Pr         | $\frac{\mu_r c_{p_r}}{k_r}$                     | $2.5 \times 10^{23}$ | Prandtl number                                          |
| $M^2$      | $\frac{k_r^2 \chi_{T_r}}{\rho_r c_{p_r}^2 b^2}$ | $10^{-33}$           | Mach number (squared)                                   |
| $\epsilon$ | $\alpha_r \Delta T_r$                           |                      | Fractional density change due to temperature variations |